No of Pages : 5 Course Code : 23MX12

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004

SEMESTER EXAMINATIONS, DECEMBER 2023

MCA Semester : 1

23MX12 STRUCTURED PROGRAMMING CONCEPTS

Time : 3 Hours **Maximum Marks : 100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.
3. Course Outcome : Table

Qn.1	CO1.	Qn.2	CO2.	Qn.3	CO3.	Qn.4	CO4.

1. a) Compare and Contrast Compiler and Interpreter.

   b) Explain the essences of 'Structured Programming Concepts' and its availability in C with benefits in software applications development.

   c) i) Discuss the various criteria for classification of Programming Languages. Based on these criteria fit the position of Programming Language C in the classification chart. Mention when to use and when not use C for software development. [6]

   ii) Discuss the various properties of good programming language in the perspective of programmer and language developer. Enumerate reasons for the popularity of a programming language. Give simple illustrations in C for above said points [6]

2. a) Distinguish the uses of Iteration from Recursion in software applications development with an example.

   b) Compare and contrast the following programming language constructs: [3+4+3]

   i) Literals, Constants and Variables

   ii) Expressions and Statements

   iii) Sequence controls and Sequence points in Program execution

   c) i) 1) Write a C program to find all Saddle points of a given matrix along with their position details.

   - For the matrix M1, the (3, 1) cell is a saddle point because 7 is the smallest value in row 3 and the largest value in column 1.
   - For the matrix M2, the (2, 2) cell is a saddle point because 5 is the smallest value in row 2 and the largest value in column 2.
   - The matrix M3 does not have a saddle point. No value is both the smallest in its row and largest in its column.
   -

M1		
1	2	3
4	5	6
7	8	9

M2		
9	1	2
8	5	7
3	4	6

M3		
8	1	9
7	2	6
3	4	5

[6]

2) Write an in-place C program that accepts a two dimensional array A of integers represented as int A[m][n], rotate it 90 degrees clockwise. (In-place means minimal extra memory to be used, i.e., doesn't make a new array to copy into) [6]

Input

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Output

13	9	5	1
14	10	6	2
15	11	7	3
16	12	8	4

[OR]

ii) 1) Suppose in a C program, three strings PSG, Coimbatore, and Programming have to be stored. Calculate the memory requirement in bytes for each of the following two options of storing the strings in a computer with 64-bit addresses: [2+3]

Option-1: a statically declared 2D array of characters with 10 rows and 20 columns, with each word starting at a new row of the 2D array.

Option-2: a dynamically declared array of length three, where an array entry is a character pointer containing the starting address of a string.

2) Fill in the blanks for the following C program that reads in the information about a group of students from the user, and then prints the longest name among the group.

```
#include <stdio.h>
#include <string.h>
#define CLASS_SIZE 10
#define MAX_LEN 20

typedef struct
        {
         char name[MAX_LEN];
         int age;
         float height; // in cm
        } student;

_______ findlongestname(_____________________ class[]) // 2 marks
{
         int i, maxlen = 0, currlen, index;
         for (i = 0; i < CLASS_SIZE; i++)
         {
           if ((currlen = strlen(_____________________)) > maxlen) // 1 mark

                 {
                         maxlen = currlen; index = ___________; // 1 mark
                 }
        }
         return _______________; // 1 mark
}

int main()
{
```

```
        student class[CLASS_SIZE];
        int i, index;
        for (i = 0; i < CLASS_SIZE; i++)
        {
                printf("\nEnter name: ");
                scanf(" %[^\n]", class[i].name);
                printf("\nEnter age: ");
                scanf(" %d", &class[i].age);
                printf("\nEnter height: ");
                scanf(" %f", &class[i].height);
        }
        index = findlongestname(class);
        printf("\nThe longest name is: %s", class[index].name);
        return 0;
}
```

3. a) What is the output of the following C code snippet?

```
#include<stdio.h>
 #include<stdio.h>
int main()
  {
        char *arr[] = {"ant", "bat", "cat", "dog", "egg", "fly"};
        char *ptr1, *ptr2;
        int i;
        unsigned int k;
        ptr1 = (arr+1)[2];
        ptr2 = (*arr+2);
        for(i=0; i<10; i++)
        {
                k = k + (ptr2 - ptr1);
                k = k%4;
        }
      printf("Out1: %s || Out2: %s || Out3: %s\n", ptr1, ptr2, arr[k]);
      return 0;
  }
```

b) i) In this C declaration "long **(*(*(*z)(void))[7])(void); ", What is z? Give syntax for the use of 'z'. [3]

ii) You are given an m × n array A of integers, each column of which is a sorted list of size m. Your task is to merge the n sorted lists and store the merged list in a one-dimensional array B. It is given that each column does not contain repetition(s) of integers, that is, the m integers in each sorted list are distinct from one another. However, integers may be repeated in different rows. During the merging step, you must remove all these repetitions. Use dynamic memory allocated arrays and

pointer arithmetic operations for processing. Print the arrays before and after merging operation. [7]

Input A	1	2	3	4					
	3	5	6	7					
	5	7	8	9					
Output B									
	1	3	5	2	7	6	8	4	9

c) i) Discuss the following aspects of structured programming with illustrations from C.

a. Call by Value and Call by Reference [4]

b. Types of Storage Classes [5]

c. Function Pointers and its uses [3]

[OR]

ii) 1) Give the uses of 'extern', 'volatile' and 'const' keywords [4]

2) Given an array, the reverse Array function tries to reverse it using pointers. Fill up the blanks in the below code to achieve the functionality.

Example:

Input Array is {-1, 5, -8, -4, 5, 3}

Output is {3, 5, -4, -8, 5, -1} [8]

```
#include<stdio.h>
 #include<stdio.h>
// Function to swap two memory   contents

void swap(int *a, int *b)
{
int temp = *a;

--------  (1)
*b = temp;
}

void reverseArray(int array[], int array_size)
{
     // leftPointer pointing at the beginning of the array
      int *leftPointer = ----------    (2)     ;

    // rightPointer pointing at end of the array
      int *rightPointer = ----------------    (3)     ;

    while (---------(4)        )
      {
         swap(leftPointer, rightPointer);
          --------      (5)     ;
           --------     (6)     ;
      }
}
```

4. a) How to decide when to use and when not use Patterns and Frameworks for application development?

   b) Compare and Contrast the features of System programming Languages and scripting languages in detail. Mention when to use each type of Programming Language.

   c) i) Explain the features offered by Static library, Dynamic library and Low-level Programming for application development in C. [8]

      ii) Discuss the role of GIT like tools in Source code Management aspects of any Web Application. [4]

/END/

FD/RL